Local inference

#2 AI coding meetup, Graz

Thomas Aglassinger, Providens Analytics GmbH, 2026-08-17

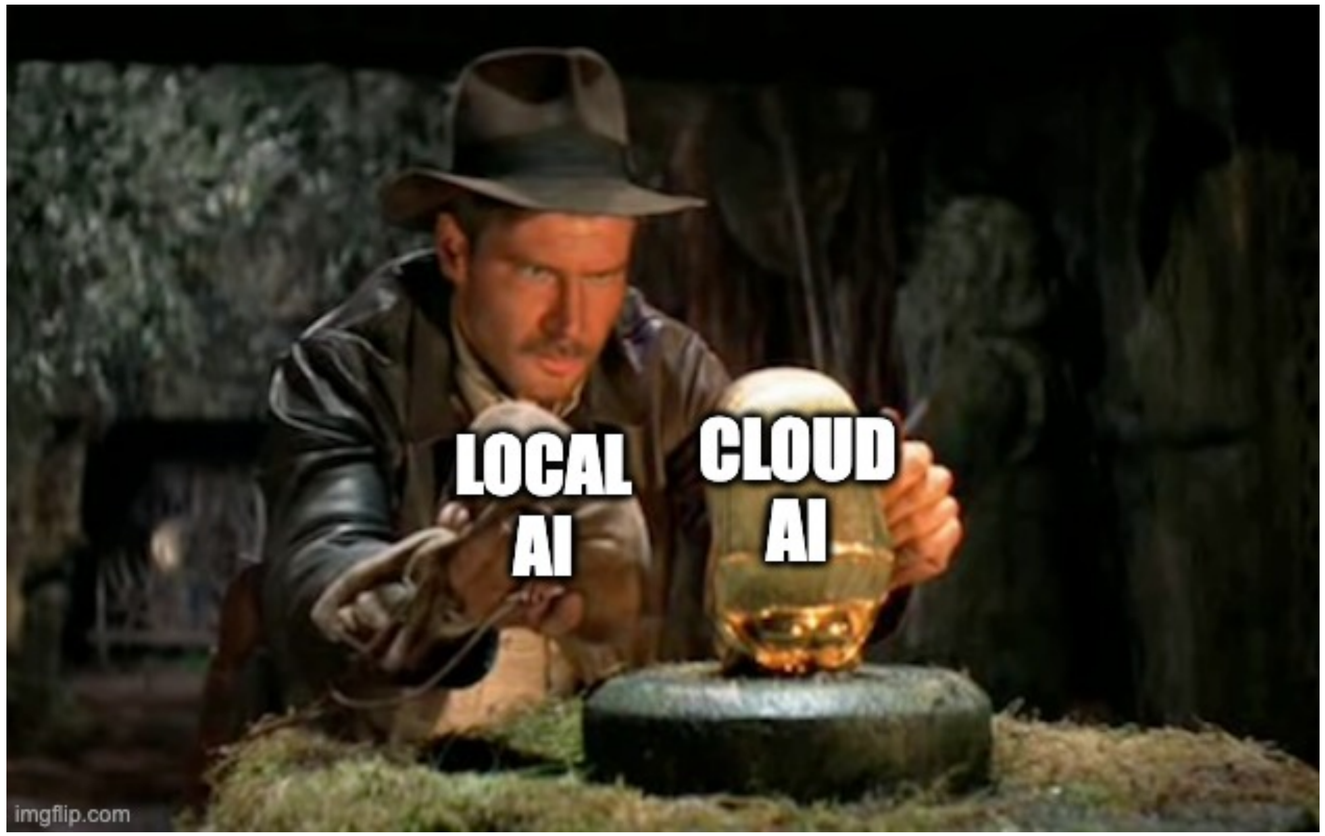
About me

Thomas Aglassinger

- Software developer and requirements engineer
- 25 years industry experience in various sectors (finance, health, e-commerce, ...)
- Currently at Providens Analytics GmbH, Graz
  - Individual solutions, system integration, strategy support
  - Focus on data, processes, workflow
  - LLMs are one tool in many we use for that
- Co-organizer of Python user group Graz

Providens Analytics GmbH
https://www.providens.at/

Common fear



Peaceful coexistence



Agenda

- Why run LLMs locally?
- Common hardware options
- Local LLM servers
- Popular models
- Local tokens per second
- Local EUR / 1M Tokens

Why run LLMs locally?

Data sovereignty

- All your data, source code, skills, and knowledge remain on premise.
- Can not be used by AI provider for training.



https://pixabay.com/photos/house-home-architecture-building-4858362/

Contractual requirement

- Certain sectors have legal requirements that all data remain on premise.
  For example, finance, health, defense.

- Customers can require that their data and trade secrets remain on premise.



Image by Andreas Breitling from Pixabay

Privacy

No surprises

Your Claude Chats May Have Been Exposed on Google

This is the second time convos with the AI have leaked onto Google Search.

Jake Peterson July 27, 2026

https://lifehacker.com/tech/your-claude-chats-may-have-been-exposed-on-google

"YIKES"

Oddest ChatGPT leaks yet: Cringey chat logs found in Google analytics tool

ChatGPT leaks seem to confirm OpenAI scrapes Google, expert says.

ASHLEY BELANGER – 7 NOV 2025 17:49 | 45



Credit: Aurich Lawson | Getty Images

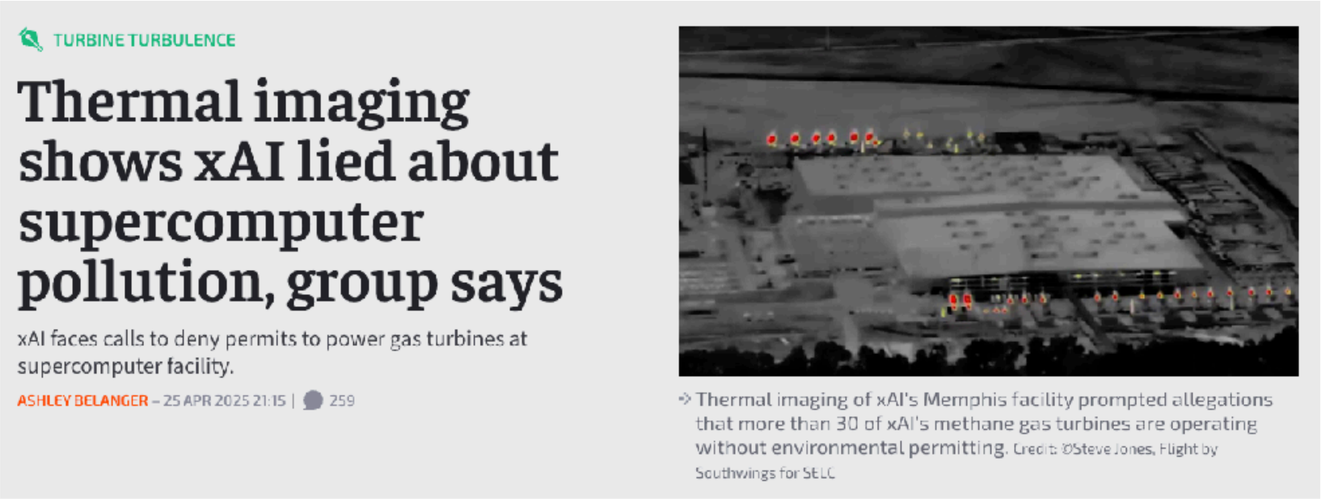
https://arstechnica.com/tech-policy/2025/11/oddest-chatgpt-leaks-yet-cringey-chat-logs-found-in-google-analytics-tool/

Sustainable energy

🏡 **Home**: Control your energy

- Use local sustainable energy provider.
- Use sustainable energy at home, e.g. solar panels.

🌩️ **Cloud**: Accept data center energy provider



TURBINE TURBULENCE

Thermal imaging shows xAI lied about supercomputer pollution, group says

xAI faces calls to deny permits to power gas turbines at supercomputer facility.

ASHLEY BELANGER – 25 APR 2025 21:15 | 259

Thermal imaging of xAI's Memphis facility prompted allegations that more than 30 of xAI's methane gas turbines are operating without environmental permitting. Credit: ©Steve Jones, Flight by Southwings for SELC

https://arstechnica.com/tech-policy/2025/04/elon-musks-xai-accused-of-lying-to-black-communities-about-harmful-pollution/

Unrestricted models



Availability

- Works without internet.
- Works if external provider is down.
- Works if external provider pulls your favorite model.
- No hourly/daily/weekly token limit.

Anthropic disables access to Fable 5 and Mythos 5 to comply with government directive

PUBLISHED FRI, JUN 12 2026•9:09 PM EDT | UPDATED FRI, JUN 12 2026•9:36 PM EDT

https://www.cnbc.com/2026/06/12/anthropic-disables-access-to-fable-5-and-mythos-5-to-comply-with-government-directive.html

OpenAI Confirms ChatGPT is Down Worldwide

1 minute read

Jul 25th, 2026 12:48 PM EEST | News

https://www.macobserver.com/news/chatgpt-is-down-worldwide-what-we-know-about-todays-outage/

Mental health

- No pressure to **max out subscription** tokens
- No cost when local LLM is not used
- Better developer health

Common hardware options

Any computer

- Larger models need more RAM.
- GPUs are faster then CPU.
- Shared RAM vs GPU RAM
- >=12b models are useful for simple everyday tasks
  → 16 GB RAM recommended



128 GB variants with 4 TB storage

	Framework Desktop	MacBook Pro M5	NVIDIA DGX Spark
GPU	AMD Max+ 395	Apple M5 Max 40 cores	Blackwell GB10
RAM (in GB)	128	128	128
Memory bandwith (in GB/s)	256	615	273
Storage (in TB)	4	4	4
Price (in €)	5,500	9,000	5,800
Link	https://frame.work/at/en/desktop	https://www.apple.com/at/shop/buy-mac/macbook-pro	https://marketplace.nvidia.com/en-us/enterprise/personal-ai-supercomputers/dgx-spark/

64 GB variants with 1 or 2 TB storage

	Framework Desktop	Macbook Pro M5	Macbook Pro M1 (used)
GPU	AMD Max+ 395	Apple M5 Max 32 cores	Apple M1 Max 24 cores
RAM (in GB)	64	64	64
Memory bandwith (in GB/s)	256	615	400
Storage (in TB)	2	2	1
Price (in €)	3000	5,100	starting at 1800
Link	https://frame.work/at/en/desktop	https://www.apple.com/at/shop/buy-mac/macbook-pro/	https://www.willhaben.at/

AMD/Intel PC with Nvidia GPU

Assumption: Basic PC without GPU = €1,000

	Single	Dual	Vintage
GPU	RTX 5090	2x RTX 5090	4x RTX 3080
RAM (in GB)	32	2x32 = 64	4x12 = 48
Memory bandwith (in GB/s)	256	615	400
Storage (in TB)	2	2	1
Price (in €)	1,000 + 4,500 = **5,500**	1,000 + 2 x 4,500 = **10,000**	1,000 + 4 x 500 = **3,000**
Link	https://frame.work/at/en/desktop	https://www.apple.com/at/shop/buy-mac/macbook-pro/	https://www.willhaben.at/

Local LLM servers

LM Studio

✅ Modern user interface

✅ Visual access to model options

✅ Presets for different kind of prompts

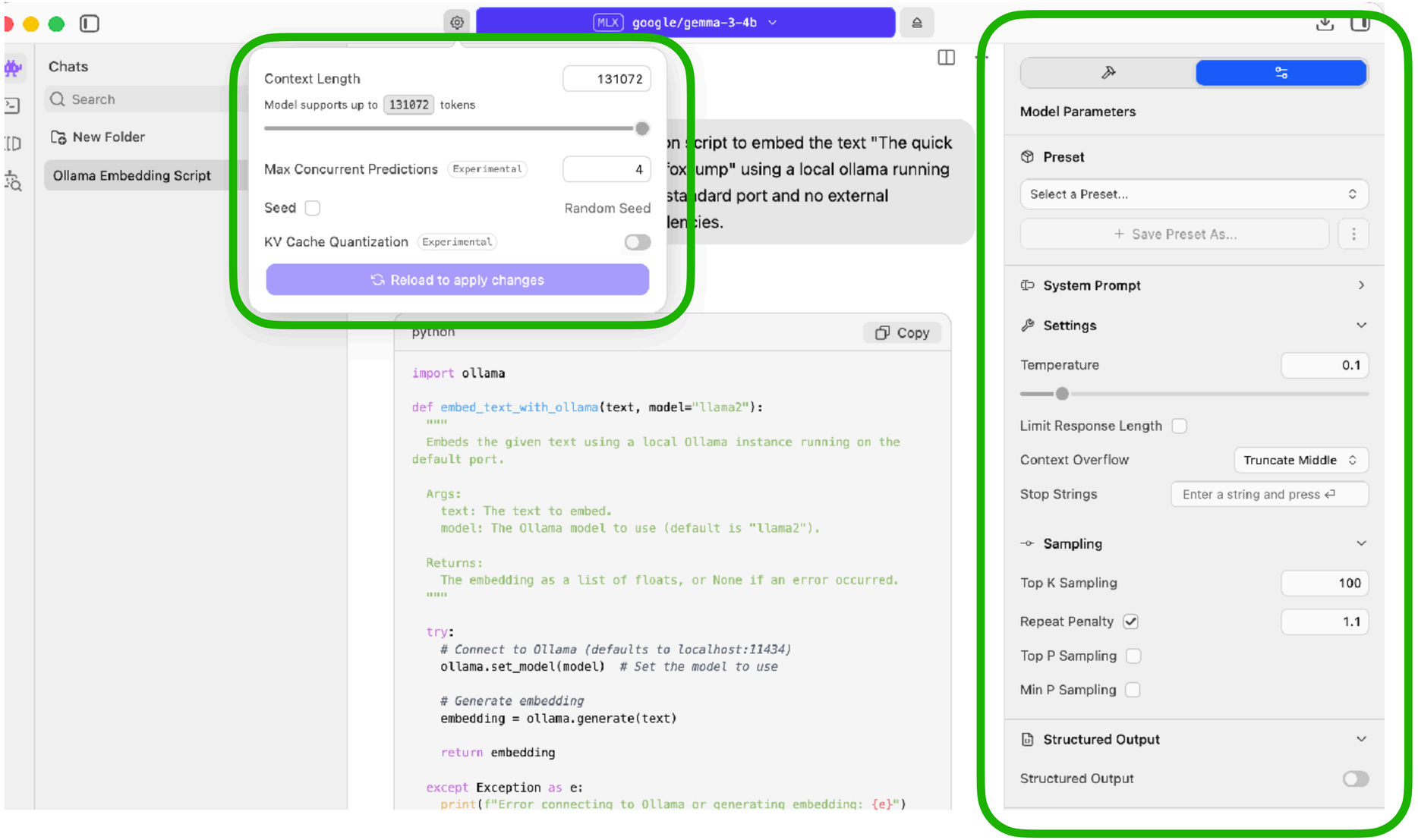
✅ Can provide server and API

✅ Good starting point for local AI

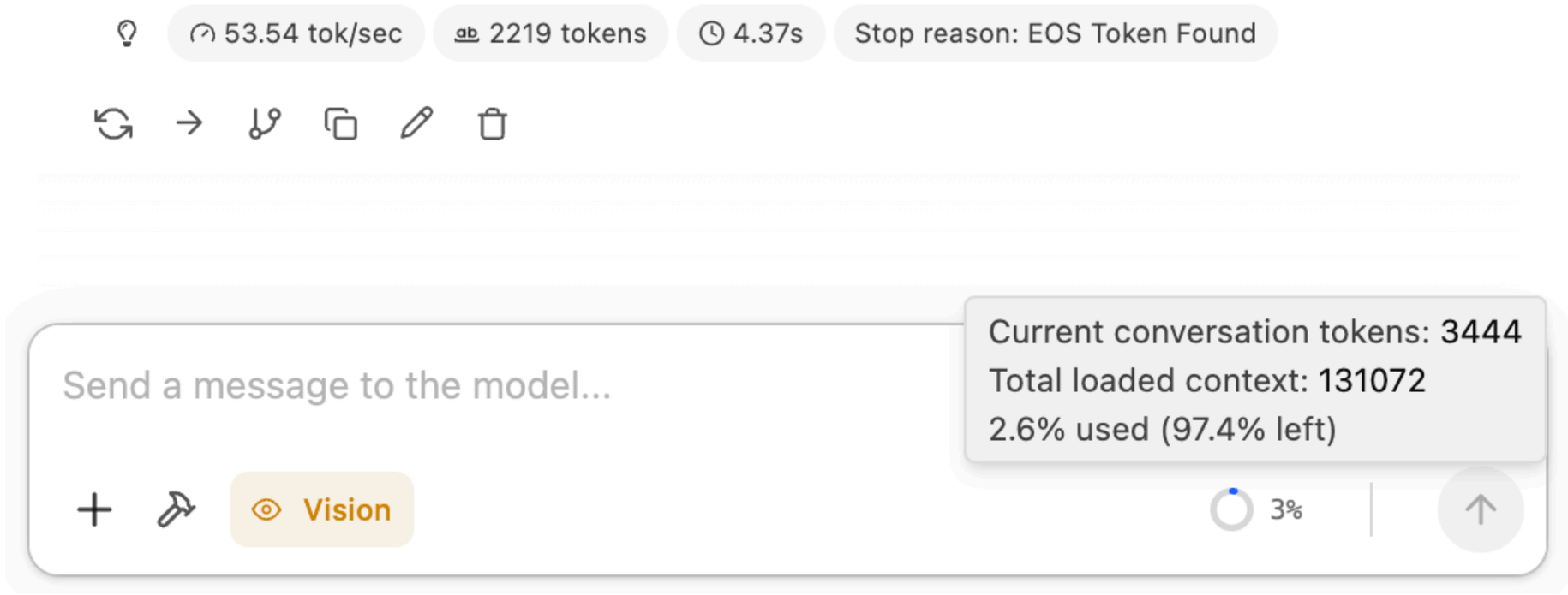
❌ Closed source

☑️ Free

LM Studio - Model settings



LM Studio - Stats at end of message



Ollama

✅ Terminal UI

✅ Server and API

✅ Homebrew integration:
brew services start ollama

✅ Juggles multiple models

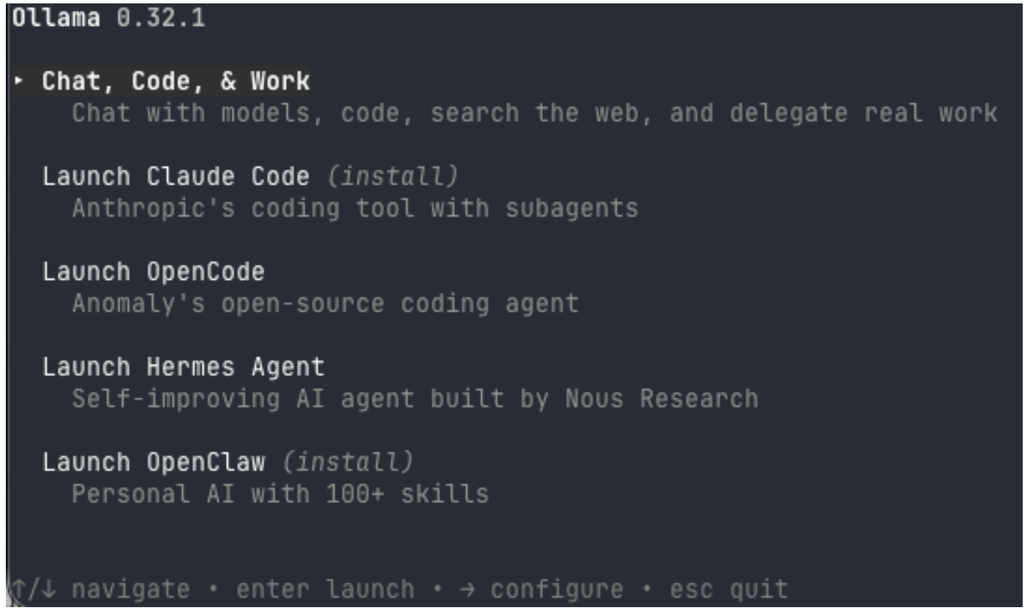
☑️ Kind-of open source

❌ Uses own model file format

❌ Uses llama.cpp but does not honor MIT license
https://github.com/ollama/ollama/issues/3185

Side note: Agentic AI with ollama launch

- See https://ollama.com/blog/launch



Ollama for agentic coding

- Be careful with large models and concurrent requests
- Default settings can easily bog down your machine
- Relevant settings:
  - OLLAMA_NUM_PARALLEL: set to 1 or 2 (default: 4)
  - OLLAMA_MAX_LOADED_MODELS: set to 1 if your model is close to memory limit; otherwise, models linger in memory
- If needed, change timeout settings of agentic harness

llama.cpp

✅ Grand daddy of open source LLM inference

✅ Foundation for LM Studio, Ollama and others

✅ Command line tool

✅ Server and API

✅ Very configurable

✅ Open source

❌ Complex

Example: See "#1 AI Coding Meetup Graz"

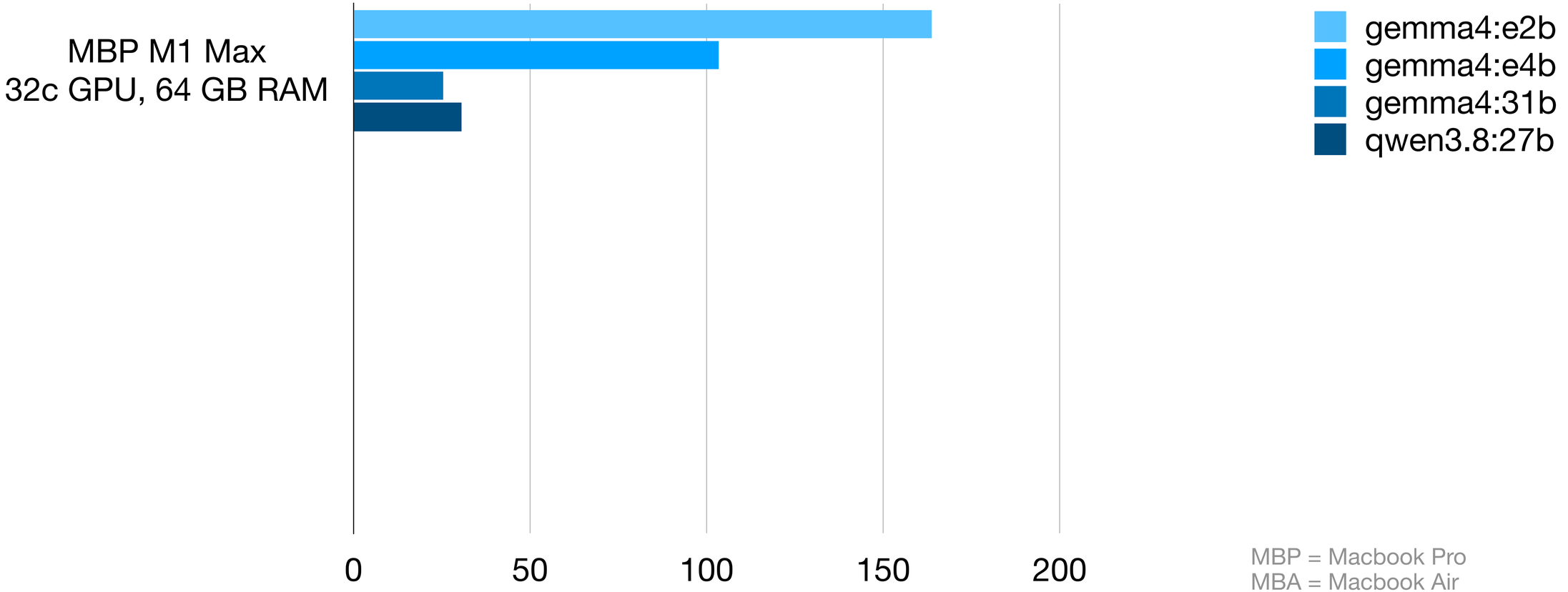
Popular models

Popular models

- With 24+GB GPU or 36+GB shared RAM:
  - qwen3.8-coding:27b
  - gemma4:30b
  - (glm-4.7-flash)
- With less RAM:
  - gemma4:e4b (comparable to 12b)
  - gptoss:20b

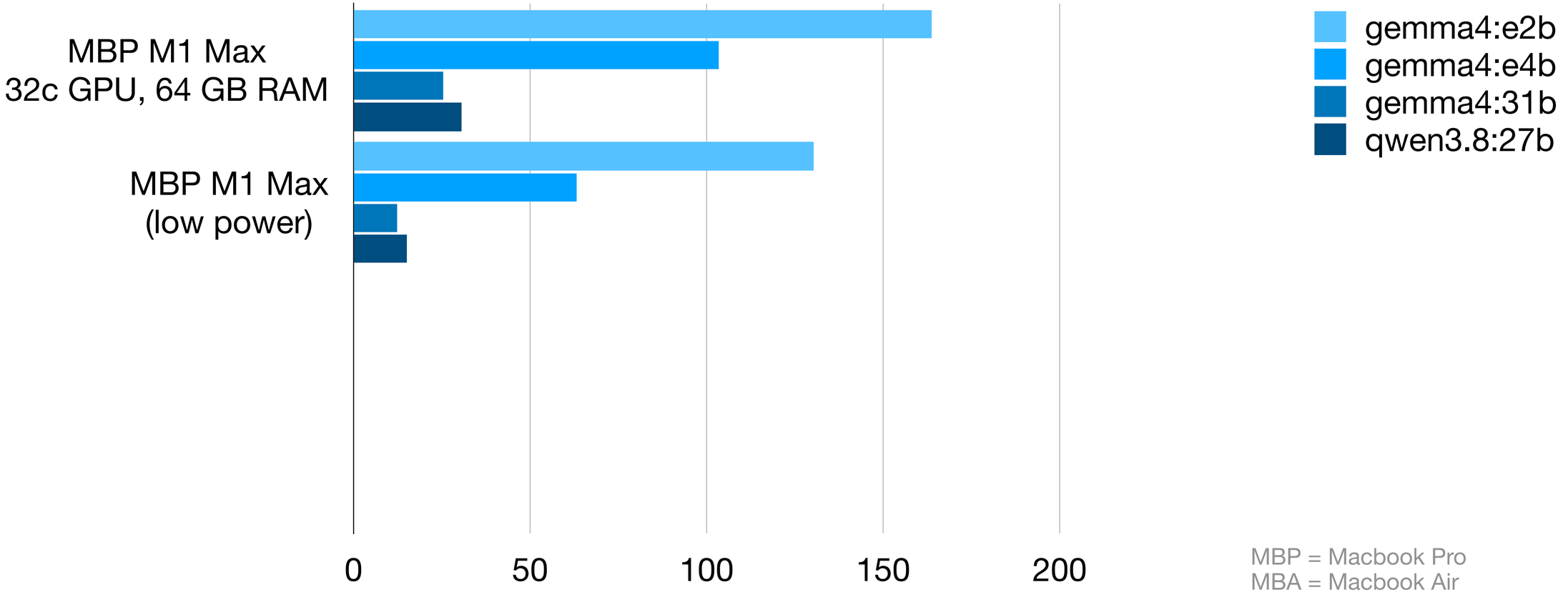
Local tokens per second

Tokens per second



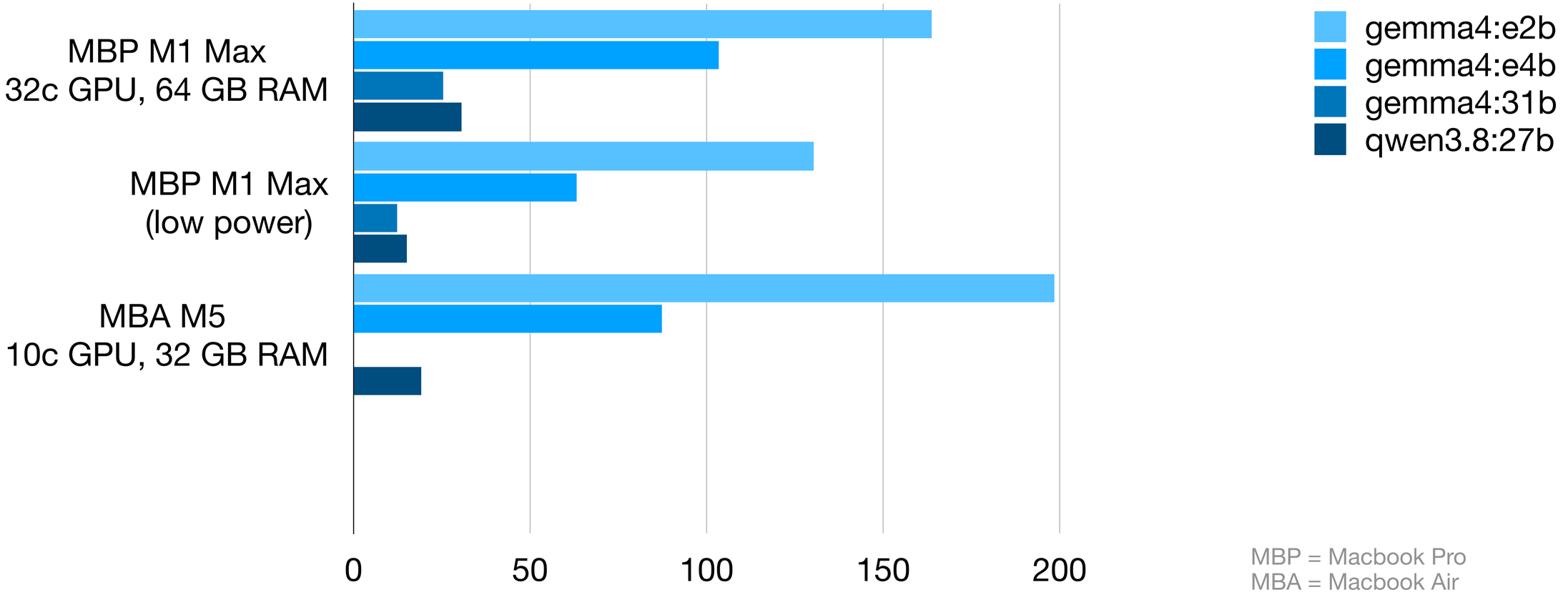
MBP = Macbook Pro
MBA = Macbook Air

Tokens per second



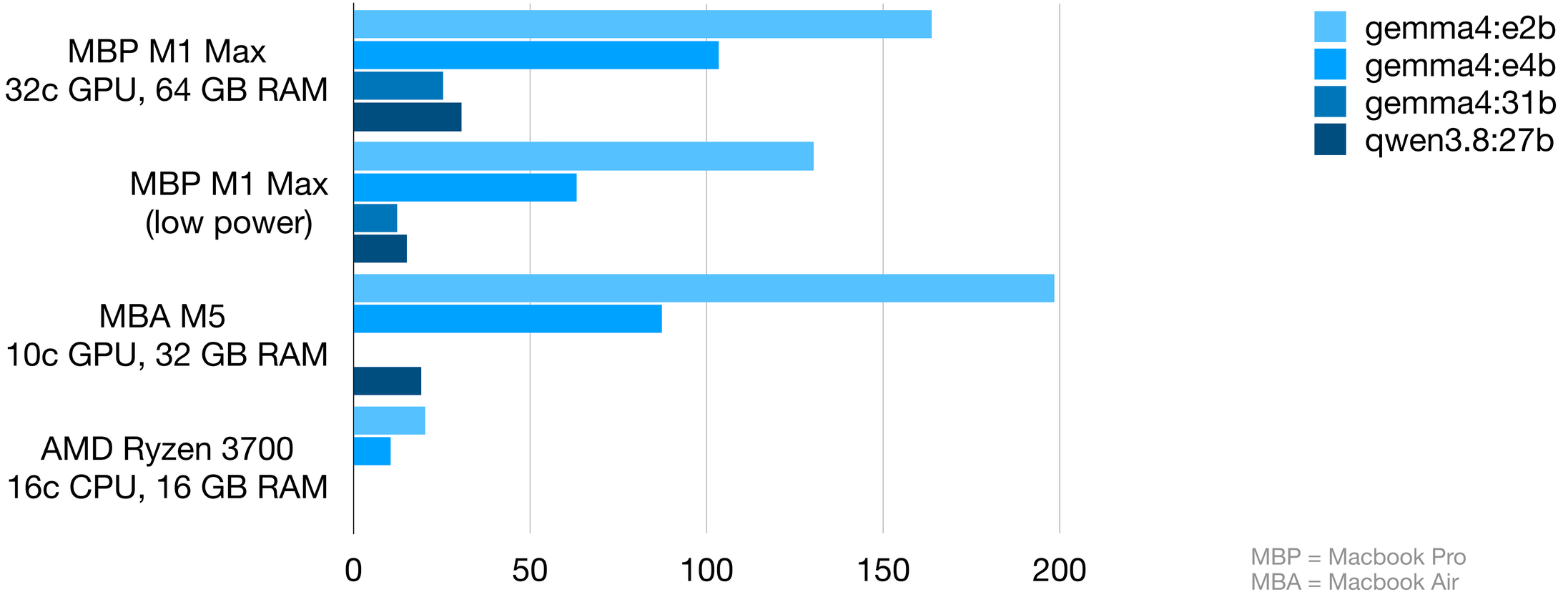
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Tokens per second



MBP = Macbook Pro
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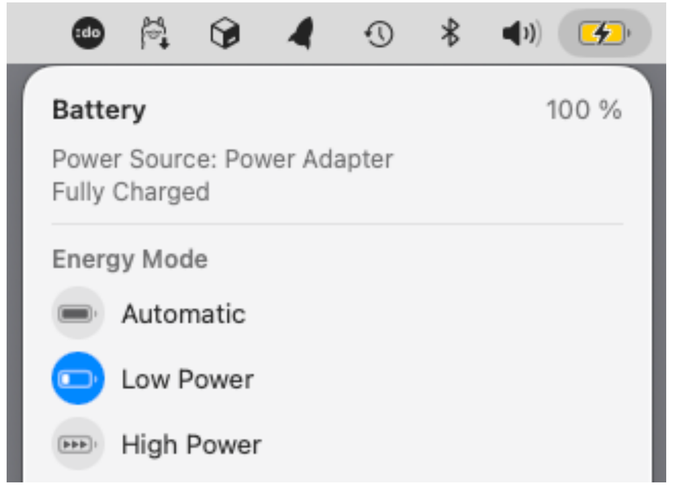
Tokens per second



MBP = Macbook Pro
MBA = Macbook Air

Noise reduction

- Put LLM server into other room.
- Use power saving mode.
- But: Slower



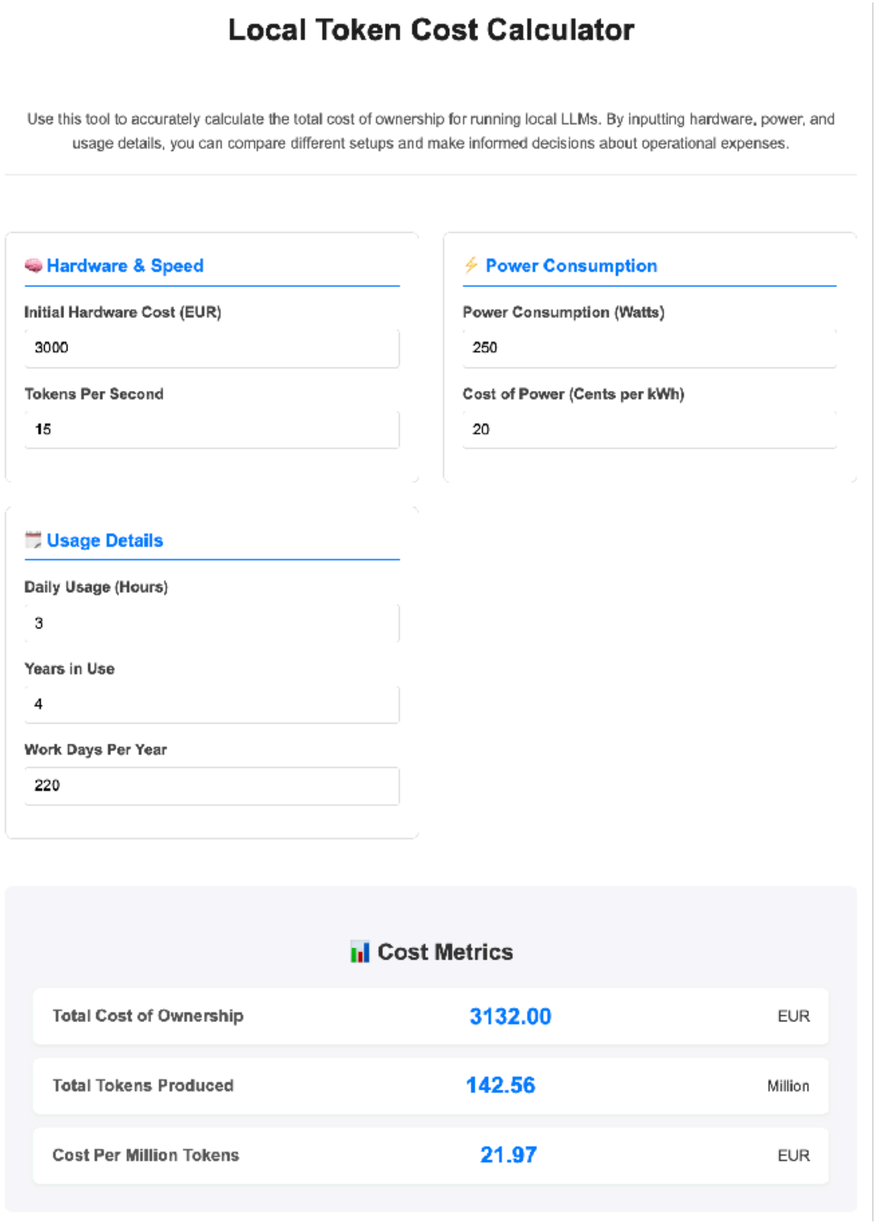
Tokens per second on your machine

- Run the demo yourself
- https://github.com/roskakori/talks/tree/master/ai_coding_graz/2026-08-17-local-inference

Local EUR / 1M Tokens

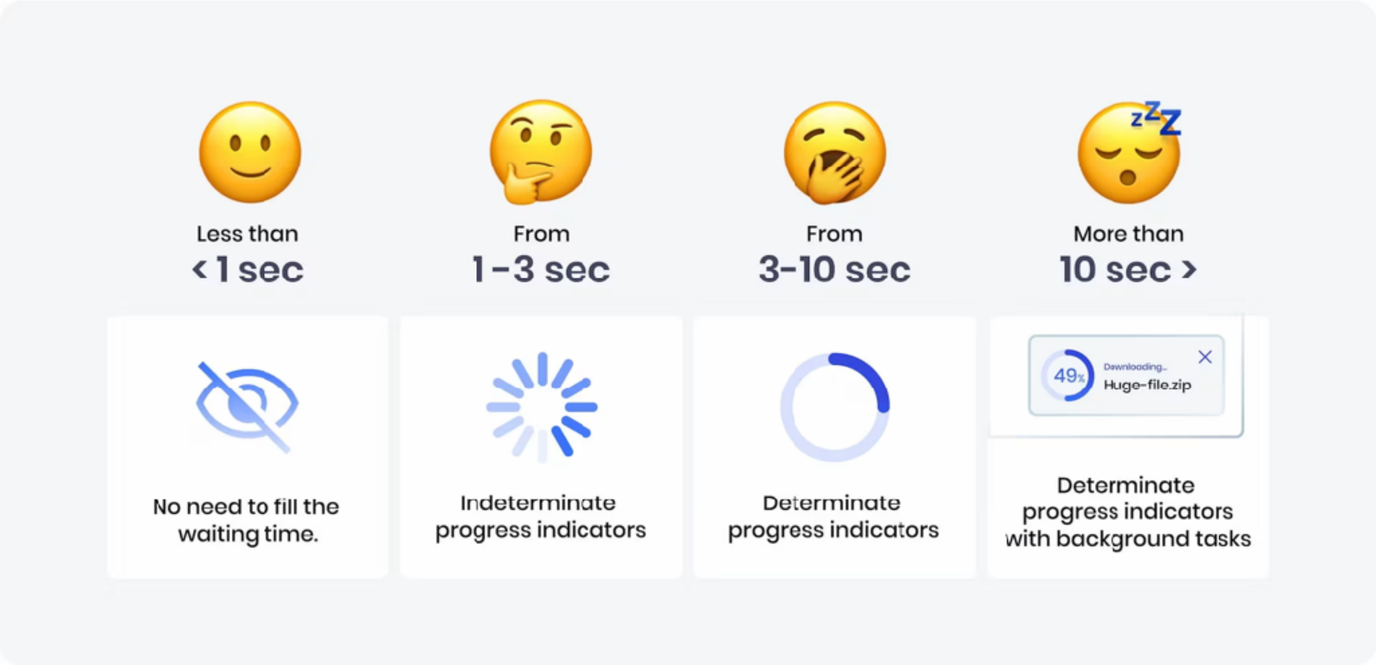
Web app

- Generated by demo using the respective model the demo performance tests
- Might or might not look like the image to the right
- Might or might not compute the same numbers



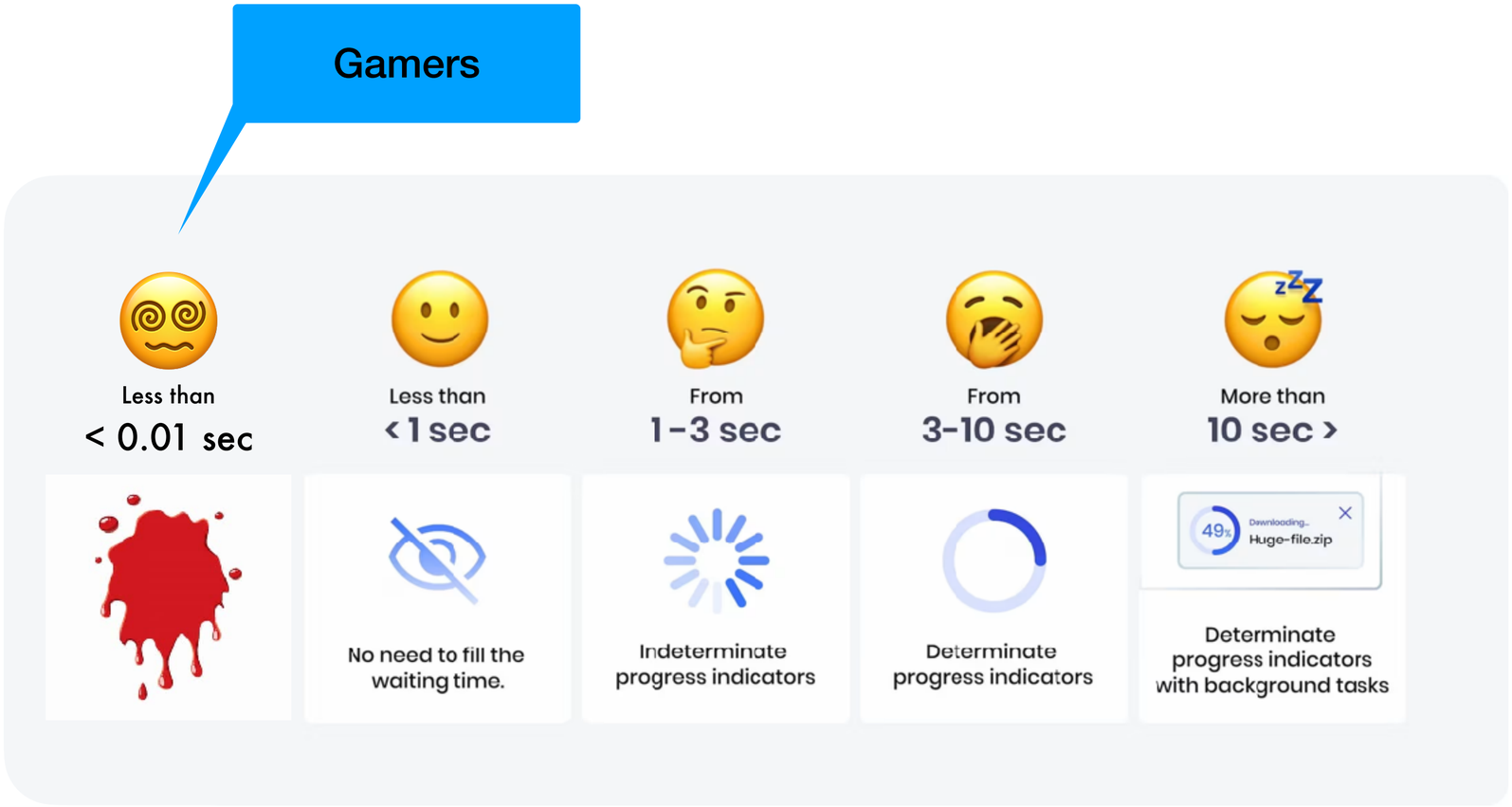
Notes on waiting times

Waiting time



https://smart-interface-design-patterns.com/articles/designing-better-loading-progress-ux/

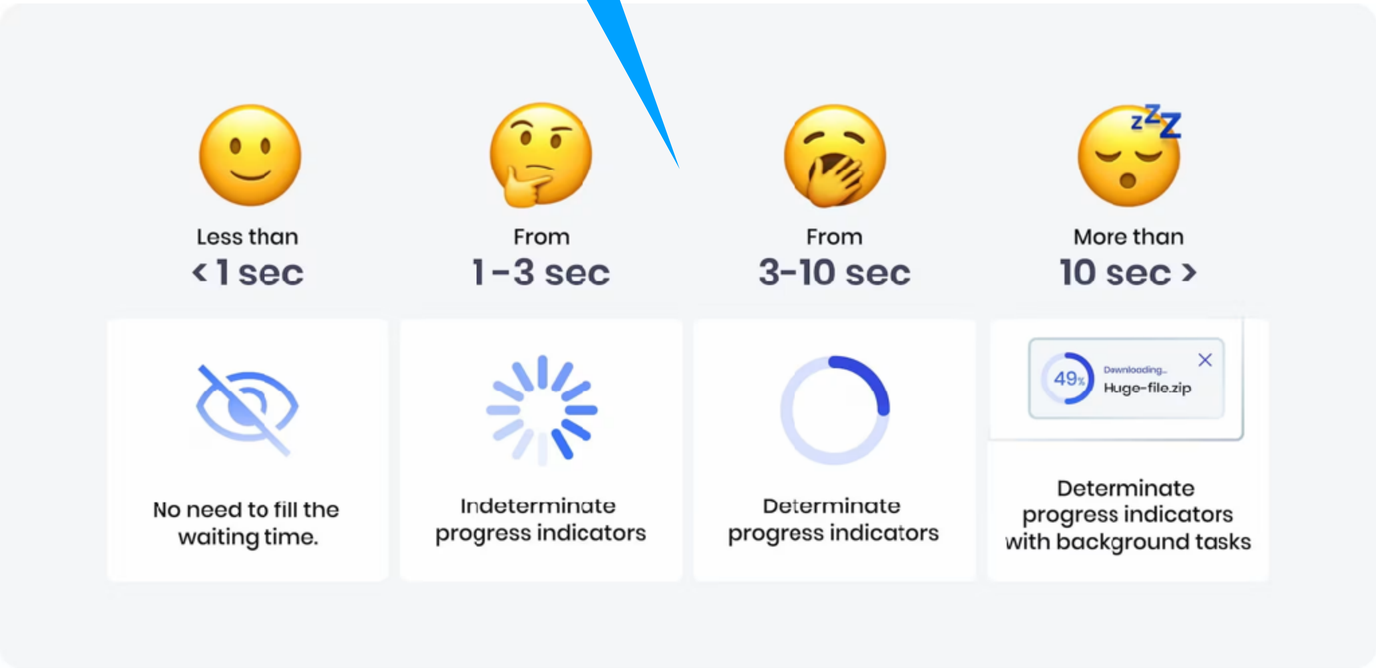
Waiting time



https://smart-interface-design-patterns.com/articles/designing-better-loading-progress-ux/

Waiting time

AI chat in cloud:
"Explain this error", "Function to do this and that", "..."

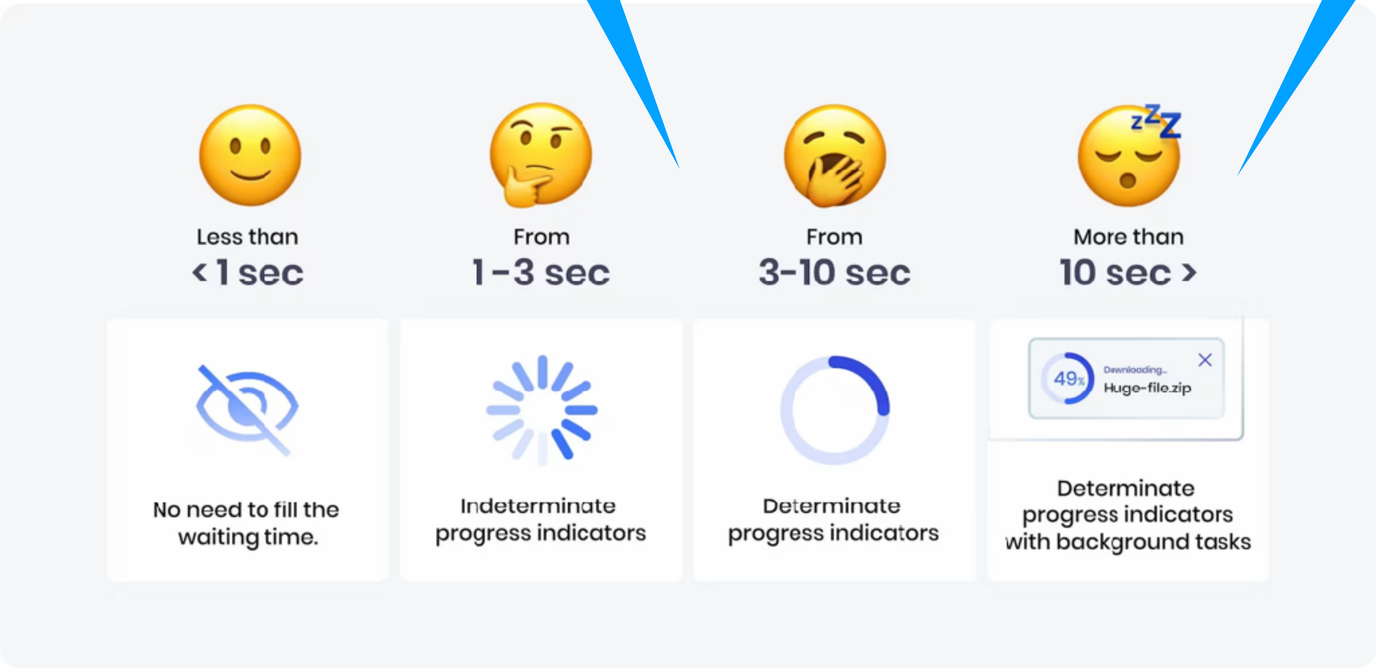


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Waiting time

AI chat in cloud:
"Explain this error", "Function to do this and that", "..."

Agentic AI in cloud and on local



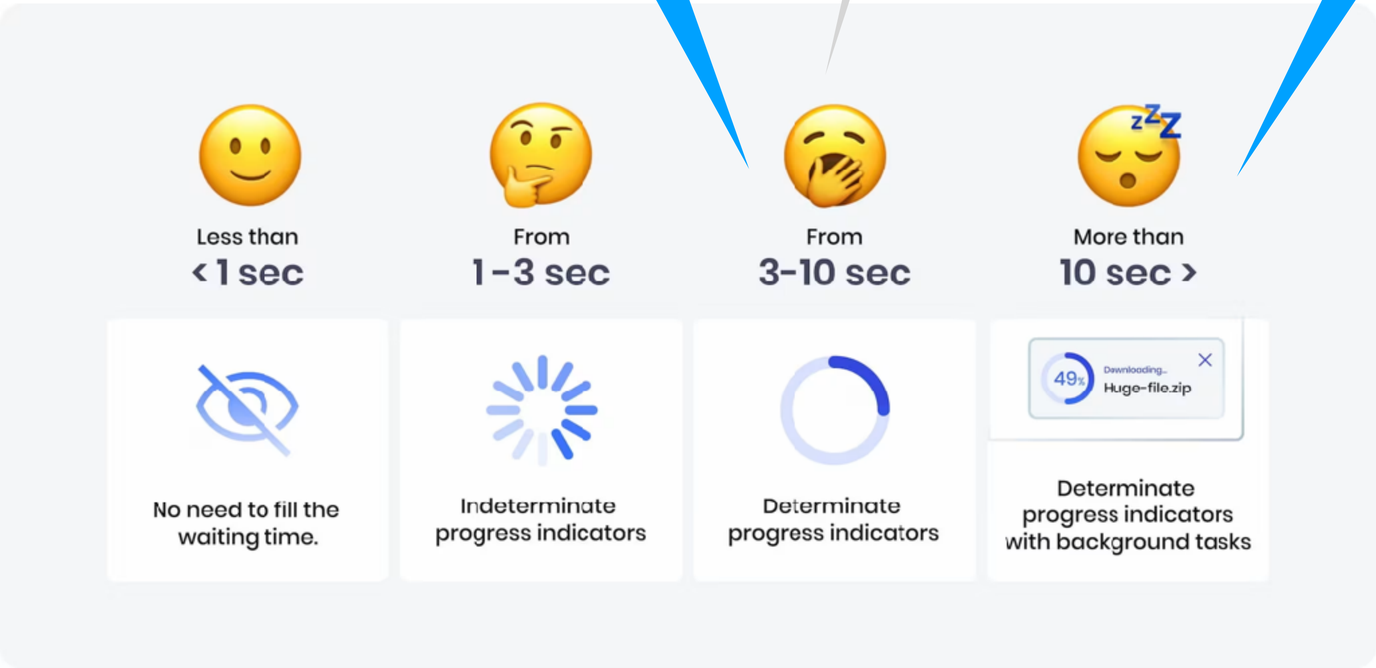
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Waiting time

super short time memory

AI chat in cloud:
"Explain this error", "Function to do this and that", "..."

Agentic AI in
cloud and on local



https://smart-interface-design-patterns.com/articles/designing-better-loading-progress-ux/

Summary

Local inference

- Local models with 20b+ work well for many tasks
- Pros: Sovereignty, privacy, sustainable, no restrictions, availability, developer mental health
- Cons: Slower, slightly less capable models, needs powerful hardware
- Mix with cloud inference as needed

Slides and code
https://github.com/roskakori/talks